Problem 5

a)

$A = T\Lambda T^{-1}$

Where:

A Is a square matrix.

T Is an invertible matrix.

Λ Is a diagonal matrix.

$$A = T\Lambda T^{-1} \tag{1}$$

Multiplying both sides of equation (1) by T from the right:

$$AT = T\Lambda \tag{2}$$

Rewriting equation (2) with T on the form $\left[t^1 ....t^n\right]$:

$$A\left[t^1 ....t^n\right] = \left[t^1 ....t^n\right]\Lambda \tag{3}$$

Equation (3) can be rewritten as follows:

$$\left[At^{(1)} ....At^{(n)}\right] = \left[\Lambda_1 t^{(1)} ....\Lambda_1 t^{(1)}\right] \tag{4}$$

Where:

$t^{(i)}$ Is the i_{th} column of T

Λ_i Is the i_{th} value of the diagonal matrix Λ

From equation (4) it can be stated that:

$$At^{(i)} = \Lambda_i t^{(i)} \tag{5}$$

From equation (5) it can be seen that $t^{(i)}$ is an Eigenvector of the matrix A and Λ_i is its corresponding Eigenvalue.

b)

$A = \boldsymbol{U\Lambda U^T}$

Where:

A Is a symmetric matrix.

$\boldsymbol{U}$ Is an orthogonal matrix.

$\boldsymbol{\Lambda}$ Is a diagonal matrix.

$$A = \boldsymbol{U\Lambda U^T} \tag{6}$$

Multiplying both sides of equation (6) by $\boldsymbol{U}$ from the right:

$$A\boldsymbol{U} = \boldsymbol{U\Lambda} \tag{7}$$

Notice $\boldsymbol{U^T U} = \boldsymbol{I}$

Rewriting equation (7) with $\boldsymbol{U}$ on the form $\left[\boldsymbol{u^1} \dots . \boldsymbol{u^n}\right]$:

$$A\left[\boldsymbol{u^1} \dots . \boldsymbol{u^n}\right] = \left[\boldsymbol{u^1} \dots . \boldsymbol{u^n}\right]\boldsymbol{\Lambda} \tag{8}$$

Equation (8) can be rewritten as follows:

$$\left[A\boldsymbol{u}^{(1)} \dots . A\boldsymbol{u}^{(n)}\right] = \left[\Lambda_1 \boldsymbol{u}^{(1)} \dots . \Lambda_1 \boldsymbol{u}^{(1)}\right] \tag{9}$$

From equation (9):

$$A\boldsymbol{u}^{(i)} = \Lambda_i \boldsymbol{u}^{(i)} \tag{10}$$

Where:

$\boldsymbol{u}^{(i)}$ Is the i_{th} column of $\boldsymbol{U}$

Λ_i Is the i_{th} value of the diagonal matrix $\boldsymbol{\Lambda}$

From equation (10) it can be seen that $u^{(i)}$ is an Eigenvector of the matrix A and Λ_i is its corresponding Eigenvalue.

c)

If A is symmetric and PSD, then A can be written on the form:

$$\boldsymbol{A} = \boldsymbol{U\Lambda U^T} \tag{11}$$

Writing the quadratic form of $\boldsymbol{A}$**:**

$$\boldsymbol{x^T A x} = \boldsymbol{x^T U \Lambda U^T x} = \boldsymbol{y^T \Lambda y} = \sum_i^n \lambda_i y_i^2 \tag{12}$$

Notice $\boldsymbol{y} = \boldsymbol{U^T x}$

Since $\boldsymbol{A}$ is PSD, then:

$$\sum_i^n \lambda_i y_i^2 \geq 0 \tag{13}$$

With y_i^2 always positive, then λ_i must be ≥ 0 for equation (13) to be true.